CONTROL ENGINEERING

1. a) The matrix A is upper triangular. Its eigenvalues are the diagonal entries: $\lambda_1 = 0.5$ and $\lambda_2 = -0.8$. Thus the stability properties of the equilibrium point do not depend on the value of a. Since $|\lambda_1| = 0.5 < 1$ and $|\lambda_2| = |-0.8| = 0.8 < 1$, the zero equilibrium point is asymptotically stable for all $a \in \mathbb{R}$. [5 marks]

b) Equilibrium points require $x(k+1) = x(k)$. Let $\bar{x} = [\bar{x}_1 \;\; \bar{x}_2]^\top$ be an equilibrium point. We need to solve:

$$\bar{x}_1 = \bar{x}_1 + \sin(\bar{x}_2)$$

$$\bar{x}_2 = \bar{x}_2 \cos(\bar{x}_1)$$

The first equation implies $\sin(\bar{x}_2) = 0$, which means $\bar{x}_2 = n\pi$ for any integer $n \in \mathbb{Z}$. The second equation can be written as $\bar{x}_2(1-\cos(\bar{x}_1)) = 0$. This equation is satisfied if $\bar{x}_2 = 0$ or if $1 - \cos(\bar{x}_1) = 0$.

i) If $\bar{x}_2 = 0$, then both equilibrium conditions are satisfied for any $\bar{x}_1$.

ii) If $1 - \cos(\bar{x}_1) = 0$, then $\bar{x}_1 = 2m\pi$ for any integer $m \in \mathbb{Z}$.

So the equilibrium points are: $(\alpha, 0)$ for any $\alpha \in \mathbb{R}$ and $(2m\pi, n\pi)$ for any integers $m, n \in \mathbb{Z}$. [5 marks]

c) The Jacobian matrix A for the linearization around $\bar{x} = (0,0)$ is:

$$A = \left[\begin{array}{cc} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{array} \right]\Bigg|_{(0,0)} = \left[\begin{array}{cc} 1 & \cos(x_2) \\ -x_2 \sin(x_1) & \cos(x_1) \end{array} \right]\Bigg|_{(0,0)}$$

Evaluating at $(0,0)$ yields

$$A = \left[\begin{array}{cc} 1 & \cos(0) \\ -0 \cdot \sin(0) & \cos(0) \end{array} \right] = \left[\begin{array}{cc} 1 & 1 \\ 0 & 1 \end{array} \right].$$

The linearized system is $\delta x(k+1) = A\delta x(k)$. The eigenvalues of A are $\lambda_1 = 1, \lambda_2 = 1$. These are on the unit circle. The minimal polynomial is $(\lambda - 1)^2$ since $A - I \neq 0$. Because there is an eigenvalue on the unit circle with multiplicity greater than one in the minimal polynomial, the linearized system is polynomially unstable.
Since both eigenvalues are on the unit circle, this is a critical case: the principle of stability in the first approximation does not allow us to conclude the stability properties of the zero equilibrium for the nonlinear system. [5 marks]

d) The reachability matrix is $\mathscr{R} = [B \;\; AB]$, namely

$$\mathscr{R} = \left[\begin{array}{cc} 1 & 1+2b \\ b & -b \end{array} \right].$$

The system is reachable if and only if $\text{rank}(\mathscr{R}) = 2$.

$$\det(\mathscr{R}) = (1)(-b) - b(1+2b) = -b - b - 2b^2 = -2b - 2b^2 = -2b(1+b).$$

The determinant is non-zero if $b \neq 0$ and $b \neq -1$. Thus, the system is reachable for all $b \in \mathbb{R}$ except $b = 0$ and $b = -1$. [5 marks]

Answers

e) The observability matrix is $\mathscr{O} = \begin{bmatrix} C \\ CA \end{bmatrix}$, namely

$$\mathscr{O} = \begin{bmatrix} c & 1 \\ 1-c & -2 \end{bmatrix}.$$

The system is observable if and only if $\text{rank}(\mathscr{O}) = 2$.

$$\det(\mathscr{O}) = c(-2) - 1(1-c) = -2c - 1 + c = -c - 1.$$

The determinant is non-zero if $c \neq -1$. Thus, the system is observable for all $c \in \mathbb{R}$ except $c = -1$. [5 marks]

f) i) Let

$$P = \begin{bmatrix} p_1 & p_2 \\ p_2 & p_3 \end{bmatrix}$$

and note that $PB = C'$ implies

$$PB = \begin{bmatrix} p_2 \\ p_3 \end{bmatrix} = C' = \begin{bmatrix} 0 \\ 2 \end{bmatrix}.$$

Then $A'P + PA$ becomes

$$\begin{bmatrix} 0 & 3 \\ -2 & 0 \end{bmatrix}\begin{bmatrix} p_1 & 0 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} p_1 & 0 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 0 & -2 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 6-2p_1 \\ 6-2p_1 & 0 \end{bmatrix},$$

which implies $p_1 = 3$. Hence, the system is passive. [5 marks]

ii) The closed-loop system is described by the equations

$$\dot{x} = (A - KBC)x = \begin{bmatrix} 0 & -2 \\ 3 & -2K \end{bmatrix} x = A_{cl}x.$$

The characteristic polynomial of the matrix A_{cl} is given by

$$\det(sI - A_{cl}) = s^2 + 2Ks + 6,$$

hence, by Routh test, its roots have negative real part if and only if $K > 0$. Hence the closed-loop system is asymptotically stable for all $K > 0$. [5 marks]

iii) Note that

$$\begin{bmatrix} \bar{s}I - A & B \\ C & 0 \end{bmatrix} = \begin{bmatrix} \bar{s} & 2 & 0 \\ -3 & \bar{s} & 1 \\ 0 & 2 & 0 \end{bmatrix}.$$

The determinant of this matrix is $2\bar{s}$. The matrix has rank less than 3 when $\bar{s} = 0$, showing that the system has a zero for $\bar{s} = 0$. [5 marks]

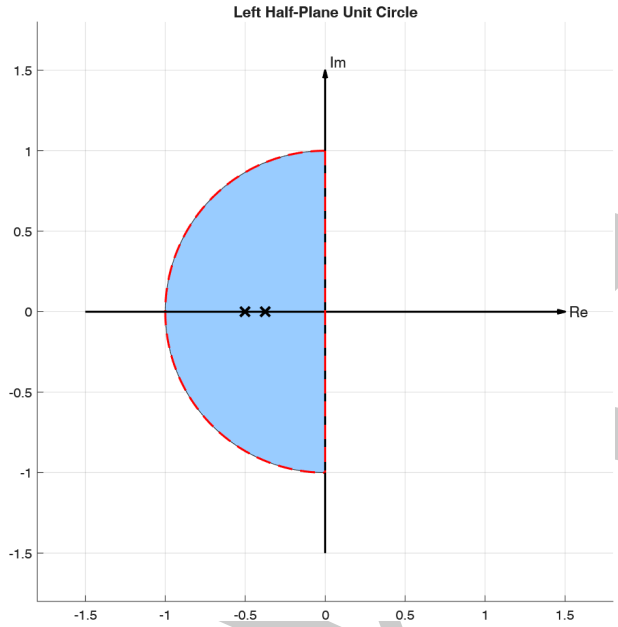
Answers

2. a) The matrix A is

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

For a continuous-time system the zero equilibrium is polynomially unstable (two eigenvalues at zero with minimal polynomial A^2). For a discrete-time system the zero equilibrium is dead-beat stable (it reaches $(0,0)$ at most in two steps). [5 marks]

b) The set of stability is the intersection of the (open) left half of the complex plane with the interior of the unity disk, that is the half disk indicated in the figure below. Note that the red points do not belong to the set.



[5 marks]

c) The indicated points do belong to the admissible region in the figure above (they are indicated with x-marks). Note that

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad K = \begin{bmatrix} k_1 & k_2 \end{bmatrix}, \qquad A_{cl} = \begin{bmatrix} 0 & 1 \\ k_1 & k_2 \end{bmatrix}.$$

The characteristic polynomial of A_{cl} is

$$p(\lambda) = \lambda^2 - k_2\lambda - k_1$$

and this is equal to the desired closed-loop characteristic polynomial

$$\left(\lambda + \frac{1}{2}\right)\left(\lambda + \frac{3}{8}\right)$$

provided $k_1 = -\dfrac{3}{16}$ and $k_2 = -\dfrac{7}{8}$. [5 marks]

d)

i) Since the system behaves like a continuous time system for $t \in [0,1)$ one has

$$x(t) = e^{A_{cl}t}x(0) \qquad t \in [0,1),$$

Answers

hence $x(1) = e^{A_{cl}}x(0)$. At $t = 1$ the system behaves like a discrete-time system, that is $\bar{x}^+ = A_{cl}x$, hence, the state is updated to

$$\bar{x}(1) = A_{cl}x(1) = A_{cl}e^{A_{cl}}x(0).$$

The state then evolves as a continuous-time system for $t \in [1,2)$ yielding

$$x(2) = e^{A_{cl}}\bar{x}(1) = e^{A_{cl}}A_{cl}e^{A_{cl}}x(0)$$

and, after one more step of the discrete-time iteration

$$\bar{x}(2) = A_{cl}e^{A_{cl}}A_{cl}e^{A_{cl}}x(0) = (A_{cl}e^{A_{cl}})^2x(0),$$

from which the general expression in the exam paper follows. In fact, the sequence $x(k)$ can be thought of as generated by a discrete-time system with "A" matrix given by

$$A_d = A_{cl}e^{A_{cl}}.$$

[5 marks]

ii) Let v be an eigenvector of A_{cl} and recall that A_{cl} has eigenvalues $\lambda_1 = -1/2$ and $\lambda_2 = -3/8$. As a result

$$e^{A_{cl}}v = e^{\lambda_i}v$$

and

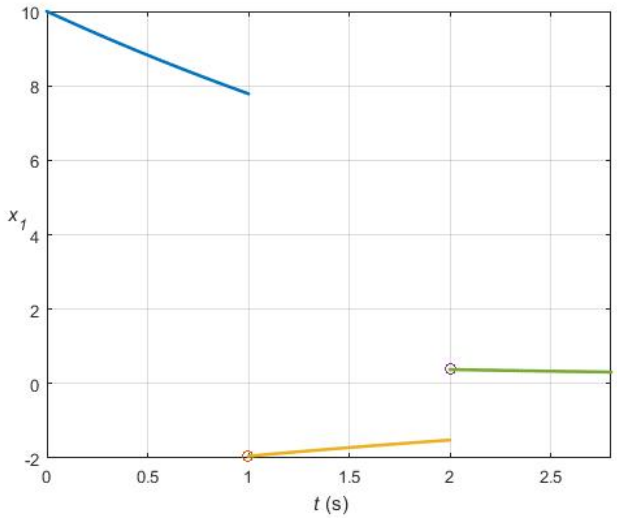
$$A_{cl}e^{A_{cl}}v = A_{cl}e^{\lambda_i}v = e^{\lambda_i}A_{cl}v = e^{\lambda_i}\lambda_i v,$$

that is the eigenvalues of $A_{cl}e^{A_{cl}}$ are

$$-\frac{1}{2}e^{-\frac{1}{2}} \approx= -0.30, \qquad -\frac{3}{8}e^{-\frac{3}{8}} \approx= -0.26.$$

Both eigenvalues are in the unity disk, hence the considered discrete-time system is asymptotically stable. [6 marks]

iii) A possible evolution of $x_1(t)$ is given in the figure below, in which the "o" sign indicate the *jumps* due to the discrete-time evolution. The change in sign at the *jumps* is due to the fact that both eigenvalues of $A_{cl}e^{A_{cl}}$ are negative.



[4 marks]

Answers

3. a) The matrices A and C are given by

$$A = \begin{bmatrix} 0 & 1 \\ \varepsilon & 2, \end{bmatrix} \qquad C = \begin{bmatrix} -\varepsilon & 1 \end{bmatrix}.$$

The characteristic polynomial is $\lambda^2 - 2\lambda - \varepsilon$, so the eigenvalues are $1 \pm \sqrt{1+\varepsilon}$. [4 marks]

b) If $1+\varepsilon > 0$, then $1+\sqrt{1+\varepsilon} > 1$ because this is 1 plus a positive number, thus implying that the system is unstable. If $1+\varepsilon < 0$ then $|1+\sqrt{1+\varepsilon}| = \sqrt{1^2 - 1 - \varepsilon} = \sqrt{-\varepsilon} > 1$, thus implying that the system is unstable. If $1+\varepsilon = 0$, then the system has two eigenvalues at $\lambda_1 = \lambda_2 = 1$. The characteristic polynomial is $(\lambda-1)^2$. Since $A - I \neq 0$, the multiplicity of the eigenvalue in the minimal polynomial is greater than one, and the system is unstable. [5 marks]

c) The observability matrix is

$$\mathscr{O} = \begin{bmatrix} -\varepsilon & 1 \\ \varepsilon & 2-\varepsilon \end{bmatrix}.$$

Note that $\det \mathscr{O} = \varepsilon^2 - 3\varepsilon$, hence the system is observable for all $\varepsilon \neq 0$ and $\varepsilon \neq 3$. The unobservable modes can be computed using the PBH test: for $\varepsilon = 0$, the two eigenvalues are 0 and 2, and the unobservable mode is at $\lambda = 0$, whereas for $\varepsilon = 3$ the two eigenvalues are -1 and 3, and the unobservable mode is $\lambda = 3$. [6 marks]

d) The system is reconstructable, by observability, for all $\varepsilon \neq 0$ and $\varepsilon \neq 3$. For $\varepsilon = 0$, since the unobservable mode is at $z = 0$ it is reconstructable, whereas it is not reconstructable for $\varepsilon = 3$ because the unobservable mode is not $z = 0$. (One could check reconstructability using alternative conditions, which however would require longer computations.) [5 marks]

e) Note that

$$A + LC = \begin{bmatrix} -\varepsilon l_1 & 1 + l_1 \\ \varepsilon - \varepsilon l_2 & 2 + l_2 \end{bmatrix},$$

which has the characteristic polynomial

$$\det(\lambda I - A + LC) = \lambda^2 + \lambda(\varepsilon l_1 - l_2 - 2) + (\varepsilon l_2 - 3\varepsilon l_1 - \varepsilon).$$

This should be equal to λ^2, yielding

$$l_1 = \frac{3}{\varepsilon - 3}, \qquad l_2 = \frac{\varepsilon + 6}{\varepsilon - 3}.$$

Note that

$$\lim_{\varepsilon=0} l_1 = -1 \qquad \lim_{\varepsilon=0} l_2 = -2,$$

whereas l_1 and l_2 are not defined for $\varepsilon = 3$. This is consistent with the fact that for $\varepsilon = 0$ the unobservable mode is at $\lambda = 0$, that is it coincides with one of the desired eigenvalues, whereas for $\varepsilon = 3$ the unobservable mode does not coincide with one of the desired eigenvalues. [6 marks]

f) The matrix $A + LC$ with $L = L_0$ is

$$A + L_0 C = \begin{bmatrix} \varepsilon & 0 \\ 3\varepsilon & 0 \end{bmatrix},$$

which has eigenvalues equal to 0 and ε. Hence the gain L_0 is such that $A + LC$ is Schur for all ε such that $|\varepsilon| < 1$. [4 marks]

Answers